



NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2009

INFORMATION TECHNOLOGY: PAPER I

Time: 3 hours

180 marks

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 12 pages. Please check that your question paper is complete.
 2. Read the questions carefully and make sure that you answer all parts of each question.
 3. Answer ALL questions.
 4. Questions 2.1, 3.3 and 6.3.1 must be answered on the **ANSWER SHEET** provided.
 5. Show all working where applicable.
 6. Non-programmable calculators may be used.
 7. Number your answers exactly as the questions are numbered.
 8. Start your answer to each question on a new page.
 9. Please leave a line open between questions and sub-questions.
 10. It is in your own interest to write legibly and to present your work neatly.
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For some years Nadia Loots observed how labourers would wait on street corners near housing estate entrances, hoping for a day's job doing gardening, tiling, painting or whatever. She decided to form a non-profit organisation, called Skilled Labour Source, to help these labourers find suitable jobs.

A business friend, Vusi Kekana, donated two laptops for Nadia's use. He also offered to give her general advice and technical help on a regular basis.

Nadia and her assistants used the laptops to create and manage a database of these labourers and the skills they offered. Home owners or building contractors could contact Nadia to find labourers with specific skills.

QUESTION 1

Nadia has asked Vusi to explain the following terms to her. Write down a definition of each of these computing terms.

- 1.1 eavesdropping
- 1.2 multitasking
- 1.3 checksum
- 1.4 keylogger
- 1.5 WIMAX (Worldwide Interoperability for Microwave Access)
- 1.6 Cloud computing
- 1.7 RSS feed (Really Simple Syndication feed)
- 1.8 Podcast
- 1.9 Router
- 1.10 RAID (Redundant Array of Independent Disks)

20 marks

Below are the specifications for the two laptops that Nadia received. Questions 2, 3 and 4 refer to these laptops.

TRAVELMATE	EMACHINE
Acer TravelMate 6293-844G32Mn Centrino Intel Core 2 Duo P8400 2.26Ghz 1066Mhz FSB 3MB L2 Cache 12.1" WXGA Crystalbrite (1280×800 resolution) white LED backlight TFT LCD Display Intel® GMA 4500MHD (64MB Dedicated/ 1695MB Shared) graphics 4096MB DDR3-1033 SO-DIMM Memory (2 × 2048) 320GB SATA 7200rpm Hard Drive DVD super multi writer optical drive 802.11a/b/g/n wireless LAN Gigabit Ethernet LAN 6 cell battery 0.3MP Acer CrystalEye Webcam Bluetooth, Fingerprint Sensor, 3G built in Microsoft Windows Vista Business Edition Windows XP Pro CD's included	eMachine eME510 Notebook Intel Celeron M 560 2.26GHz 533Mhz FSB 1MB L2 Cache 15.4" WXGA Crystalbrite (1280x800 resolution) TFT LCD Display Intel X3100 358MB graphics (8MB Dedicated/ 350MB Shared) 512MB DDR2-667 SO-DIMM Memory (1 × 512) 80GB SATA Hard Drive DVD writer optical drive 802.11g wireless LAN 10/100 Ethernet LAN 6 cell battery Microsoft Windows Vista Home Basic edition Microsoft Works No modem, No PCMCIA port No PCI-Express port, No card reader

QUESTION 2

- 2.1 Fill in Table 1 on the **ANSWER SHEET** so that it will show the specifications for the two laptops according to the following headings:
- the amount of level 2 cache
 - the speed of the system bus
 - the amount of RAM
- (6)
- 2.2 The Travelmate has a fingerprint sensor (see 3rd last line of the specification). What is the advantage of this fingerprint sensor?
- (2)
- 2.3 The Celeron processor in the eMachine only has one core. Explain how the dual core processor in the Travelmate results in a far better performance than that of the eMachine.
- (3)
- 2.4 Apart from multicore technology, name FOUR other methods that have been used over the years to improve the performance of a CPU.
- (4)
- 2.5 It is obvious that, compared to the eMachine, the Travelmate would give a far better performance for programs with a high graphics content (such as photo editing, gaming, video editing). Explain in detail why this is so, referring to each of the following (in any order):
- CPU
 - graphics card
 - system bus
- (6)

21 marks

QUESTION 3

3.1 One of Nadia's labourers, Thomas, wants to start his own paving company. He wants to use a computer to work out costs, and to create invoices.

3.1.1 Thomas manages to buy an old secondhand computer, with no software on it. He could try to obtain a cheap version of Windows, or could use a free operating system. What would you recommend to Thomas? Explain your recommendation clearly. (2)

3.1.2 Give the name of a free office suite which Thomas could use on his computer. (1)

3.2 Nadia wants to purchase a cellphone which she needs to plug into the Travelmate on a regular basis, in order to synchronise her contacts and e-mail messages.

3.2.1 Name TWO operating systems which can be used on cellphones. (2)

3.2.2 What is meant by 'synchronising' in this context? (2)

3.3 Consider the following:

Nadia downloads an email message, opens and reads it, and prints it.

On the **ANSWER SHEET**, explain the role of each of the following systems in this process:

- spam blocking software
 - anti-virus software
 - email software
 - operating system
 - BIOS
- (5)

3.4

3.4.1 One of the laptops comes with some application software listed in the specifications. Name the laptop and give the name of the application software. (2)

3.4.2 Name any TWO different **categories** of application software. (2)

3.4.3 Name ONE **category** of system software other than an operating system. (1)

3.5 Two functions of an operating system are:

- process management
- memory management

Name TWO other functions of an operating system. (2)

3.6 Explain how virtual memory works. (3)

22 marks

QUESTION 4

- 4.1 Both laptops can be used in networks which are based on Ethernet.
- 4.1.1 Name TWO types of cables that are commonly used in an Ethernet network. (2)
- 4.1.2 Name TWO different topologies that can be used in an Ethernet network. (2)
- 4.1.3 Draw a sketch to illustrate ONE of the topologies you named in 4.1.2. Use five computers in your sketch. (*Remember to write the name of the topology next to this sketch, otherwise your sketch cannot be assessed.*) (2)
- 4.2 Nowadays most networks use one or several switches. Explain the function of a switch. (3)
- 4.3 The Travelmate has four different means of making a connection with another computing device, each employing a different type of technology. These technologies are:
- Ethernet LAN
 - wireless LAN
 - Bluetooth
 - 3G
- 4.3.1 Explain how Nadia could be using each of the four network technologies in a day's work and leisure, by giving practical examples of situations where they could apply. (4)
- 4.3.2 Imagine that Nadia wants to set up a small network using three Travelmates, and three eMachines. Name ALL the options listed at 4.3 that could be used in this network. (REMEMBER to look at the computer specifications before answering.) (2)
- 4.3.3 Imagine that Nadia is setting up a peer-to-peer network. List FOUR advantages and/or disadvantages of a peer-to-peer network. (You need not list exactly two each of advantages and disadvantages; any four points, as long as each is either an advantage or disadvantage, will be accepted.) (4)
- 4.4 CSMA/CD is a network control protocol/media access control method.
- 4.4.1 Explain how CSMA/CD deals with collisions. (4)
- 4.4.2 Name one other network control protocol/media access control method. (1)

24 marks

QUESTION 5

- 5.1 What is web-based email? (3)
- 5.2 Nadia started providing clients with her email address, so that home owners or building contractors could contact her to hire specific labourers. She currently uses a web-based email address (nadia@yahoo.com). Give TWO other examples of web-based email addresses, besides the Yahoo type address. (2)
- 5.3 Vusi also tells Nadia that she needs a firewall to protect her computer from unauthorised access by outsiders. Describe TWO checks which a firewall does in order to provide this protection. (2)
- 5.4 Nadia banks with Capital Bank. She receives the following e-mail:

From: Capital Bank Internet Banking (customerservice@capitalbank.co.za)
Sent: 16 August 2009 3:34 PM
To: nadia@yahoo.com
Subject: Notice From Capital Bank: Important Security Message

Dear Customer

Capital Bank is doing a customer review. We need to verify your account details in order for you to continue your Internet banking.

In order to continue this verification please reply with your account number and PIN number.

Thank you
Capital Bank Customer Service

- 5.4.1 What word is used to describe this type of email? (1)
- 5.4.2 How should Nadia react to this email? Give a reason for your answer. (2)
- 5.5 Nadia receives an enquiry from a client regarding misconduct by one of the labourers. She draws up a document to send to this client, and decides to encrypt the document before sending it.
- 5.5.1 Why is it a good idea for Nadia to encrypt this document? (1)
- 5.5.2 Explain how encryption works. (4)
- 5.6 Nadia is frustrated because when she searches on the Internet she gets too many results. Give her TWO tips for improving her search technique. (2)
- 5.7 Nadia has a blog where she writes about her experiences with the casual labourers for whom she finds work.
- 5.7.1 What is a blog? (2)
- 5.7.2 Name ONE advantage that Nadia gains by having a blog. (1)

- 5.8 The following two extracts are from articles on the Internet which deal with copyright. Read the texts carefully and then answer questions 5.8.1 and 5.8.2.

Fair use

Fair use is one of the most important, and least clear-cut, limits to copyright. It permits some use of others' works even without approval. But when? Words like 'fair' or 'reasonable' cannot be precisely defined, but here are a few benchmarks.

Uses that advance public interests such as criticism, education or scholarship are favoured – particularly if little of another's work is copied. Uses that generate income or interfere with a copyright owner's income are not. Fairness also means crediting original artists or authors.

[From: <<http://www.piercelaw.edu/tfield/copynet.htm>>]

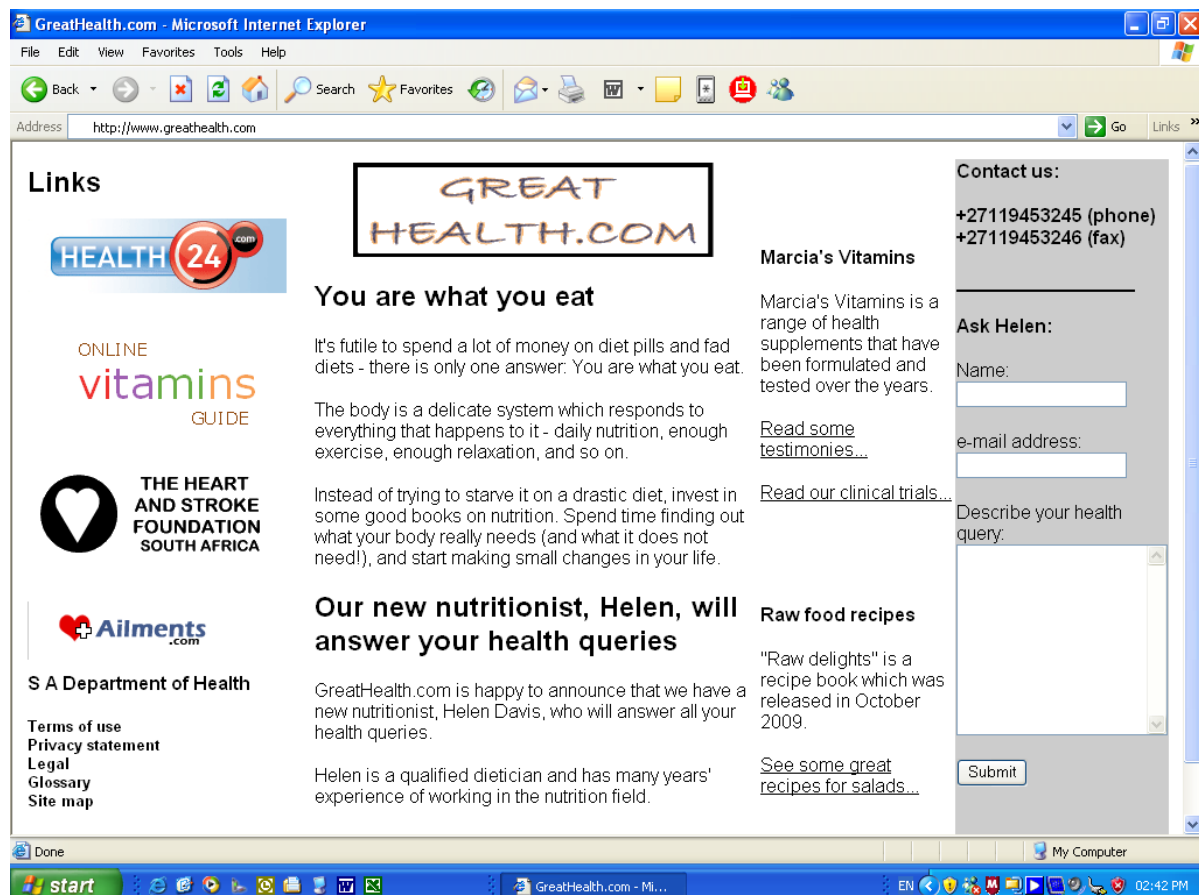
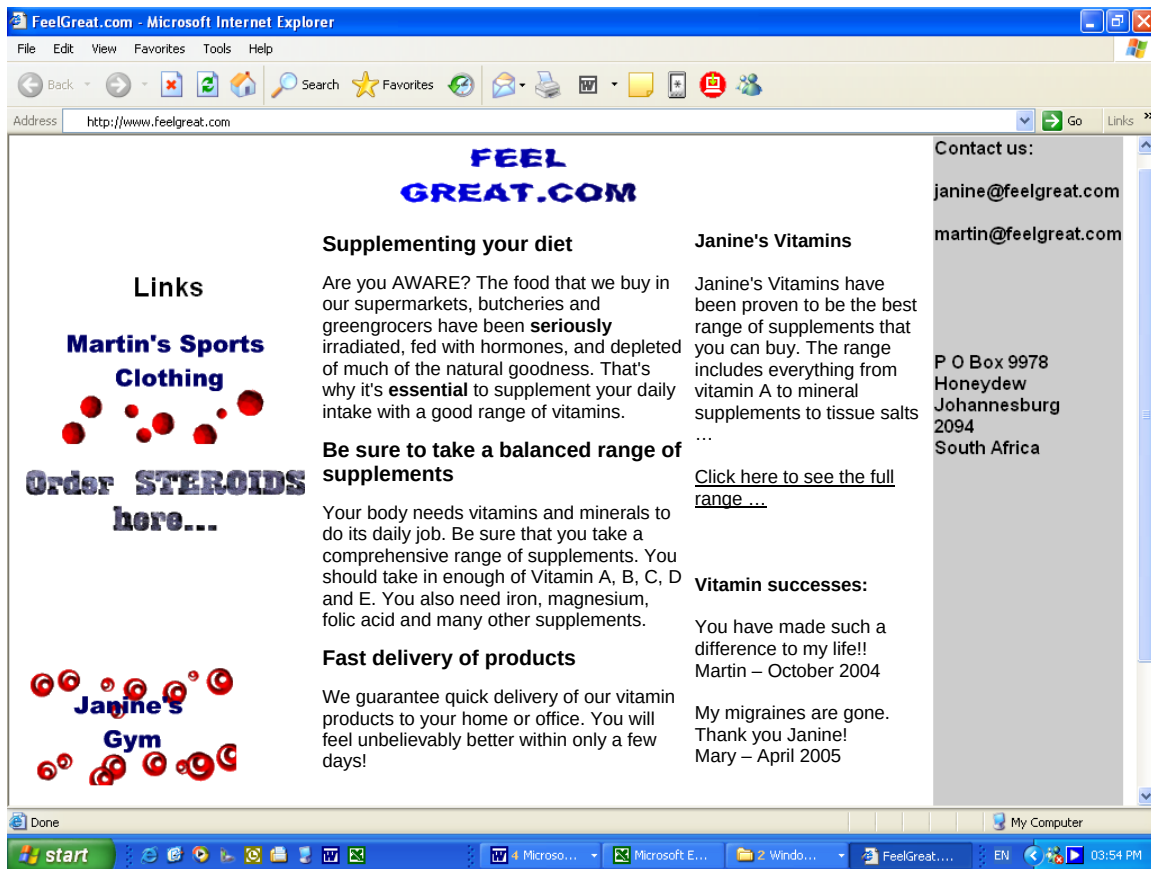
Fair Use

Fair use or fair practice is utilisation of a portion of a copyrighted work 'as is' for purposes of **parody, news reporting and research and education about such copyrighted work** without the permission of the author. Use of copyrighted works, or portions thereof, for any other purpose is not deemed fair use, so be careful! Scanning a photo of the Amazon Forest printed in *National Geographic*, and using it without permission on your personal web site will most likely not be considered as fair use. However, if you republished the photo on your site to comment on the photo as it was published in *National Geographic*, this would most likely be considered fair use. You **still have to credit your source** by naming the author of the work on the same page. If possible, take the time to contact the owner and request permission to use the owner's work.

[From: <<http://www.whatiscopyright.org/>>]

- 5.8.1 Using the **scenario of Nadia's Internet research and articles** that she writes on the employment of casual labour, give an example of an action which would be considered **not fair use**, and therefore an infringement of copyright. (2)
- 5.8.2 Again using the **same scenario**, create an example which would be considered to be **fair use**, and therefore **not** an infringement of copyright. (2)

While browsing the Internet during her free time, Nadia found the following two webpages. Study these pages and then answer questions 5.9 – 5.11.



- 5.9 The 'Great Health' site shows a privacy statement at the bottom of the page. Describe THREE kinds of protection that are usually contained in a privacy statement. (3)
- 5.10 Compare the 'Great Health' page and the 'Feel Great' page and make a judgment as to which one is a more reliable source of information on health. Motivate your answer by giving FOUR specific examples from these two webpages. (5)
- 5.11 The 'Feel Great' page shows email addresses for correspondence, whereas the 'Great Health' page does not show any email address. Instead, the 'Great Health' page provides a form on which users can submit their queries. Give one possible reason why the 'Great Health' page does not display any email addresses. (1)

33 marks

QUESTION 6

- 6.1 Name and explain TWO examples of how Nadia's labourers' lives may have improved through the use of computers by large organisations (such as the government, banks and hospitals). (4)
- 6.2 George Papadopoulos is a property developer. He builds and sells townhouses and luxury houses. Name THREE ways in which the use of computers can help him to advertise his business. (3)

(Note: Each of your answers must involve use of a different type of computer program; two ways of using the same computer program will be allocated one mark only.)

- 6.3 Joey, Michael and Thandeka are three friends who each have a job in the computer industry:

- Joey works in the Microsoft Call Centre giving online support to clients who use Microsoft Office products.
- Michael works as an IT technician for a company which provides ongoing hardware, software and networking support. He visits clients on site to attend to their problems.
- Thandeka trains students at a college in basic computer literacy, teaching them Microsoft Word, Excel and PowerPoint.

- 6.3.1 Use Table 2 on the **ANSWER SHEET** to compare the knowledge and skills required of these three people with respect to:

- (a) hardware (3)
- (b) software (3)

- 6.3.2 Imagine that Joey and Thandeka are required to swop jobs immediately, so that Joey does the computer literacy training and Thandeka works at the Microsoft Call Centre. Name TWO difficulties EACH that may be experienced by:

- (a) Joey (2)
- (b) Thandeka (2)

The networking of computers has brought many advantages to our personal and work lives.

- 6.4 Name TWO rules or guidelines which network users should follow when creating a password. (2)
- 6.5 Some companies block access to Facebook. Do you agree with this practice? Give a reason for your answer. (2)
- 6.6 Suggest TWO ways in which policing should improve with police stations being networked in a WAN across the country. (4)
- 6.7 John Smith is a client of Excellence Bank and frequently makes use of Internet banking. He finds out that a hacker has managed to remove a large amount of money from his bank account. Who is to blame – John or the bank? Discuss, explaining how both could be to blame. (4)
- 6.8 Discuss TWO ways in which the use of computers helps improve how people deal with environmental issues. (2)

31 marks

QUESTION 7

Normalisation

Nadia uses a database to keep a record of the jobs for which employers hired specific labourers. She created the following table to store the data:

tblJobs(jobID, labourer, skillID, skill, empID, employer, address, jobStartdate, jobDuration)

jobID	labourer	skillID	skill	empID	employer	address	jobStartdate	jobDuration
1	Fred	1	Bricklaying	1	Smith	Bromhof	06-Jan	2
2	Joseph	3	Tiling	4	Mbeleketi	Randburg	06-Jan	2
3	Fred	2	Painting	2	Jones	Kensington	09-Jan	3
4	Fred	1	Bricklaying	2	Jones	Kensington	21-Jan	4
5	Joseph	3	Tiling	5	Botha	Rivonia	22-Jan	5
6	Fred	1	Bricklaying	3	Mbemba	Mondeor	26-Jan	3
7	Joseph	2	Painting	5	Botha	Rivonia	7-Feb	2
8	Fred	1	Bricklaying	3	Mbemba	Mondeor	01-Feb	6
9	Martin	3	Tiling	3	Mbemba	Mondeor	16-Feb	7

- 7.1 The address field is not sufficient for storing a person's address. How would you improve the table structure to solve this? (2)
- 7.2 What is the purpose of a primary key in a database table? (1)

- 7.3 Nadia decides to make **jobID** the primary key. However, Vusi tells her: 'You can simplify this table as follows:

Delete the **jobID** field, and use **skillID** and **empID** together as a composite primary key.'

Do you agree with Vusi? Give a reason for your answer.

(Please note: No marks allocated for a 'Yes'/'No' without a reason).

(2)

- 7.4 Name TWO requirements for a table to be in 2nd normal form. (2)

- 7.5 Nadia receives notification from Mbemba that his address has changed to Alberton.

7.5.1 What do we call the anomaly that arises out of this situation? (1)

7.5.2 Explain this anomaly by referring to the details in **tblJobs**. (1)

- 7.6 What is the purpose of normalisation? (2)

SQL

One of Nadia's clients is Dudu who has a company that rents out building machinery. Dudu keeps a database table, **tblMachinery**, which shows each of his machines, along with the operator for that machine. (Each operator only works with one specific machine.)

Another table in the database (**tblRentals**) shows a list of rentals, so that Dudu can keep track of each instance where a machine was rented out, along with details of the client and the date.

Below are the first few records in each of the two tables.

tblMachinery		
macID	machine	operator
1	TLB	Khumalo
2	plate compactor	Johnston
3	TLB	Dube
4	Grinder	Tayob
5	drilling machine	Ramaboa

tblRentals			
rentID	macID	client	date
1	1	Smith	2009/09/23
2	1	Ramphele	2009/09/24
3	1	Kruger	2009/09/25
4	4	Ramphele	2009/09/27

- 7.7 Name the foreign key in **tblRentals**. (1)

- 7.8 Dudu wants to know how many machines there are of each type. He tries the following query, but it does not work.

```
SELECT machine, count(*) AS Totals
FROM tblMachinery
ORDER BY machine;
```

Write the correct SQL statement for Dudu. (2)

7.9 Dudu would like a list of all rentals, showing details as follows:

machine rented, operator, client, date

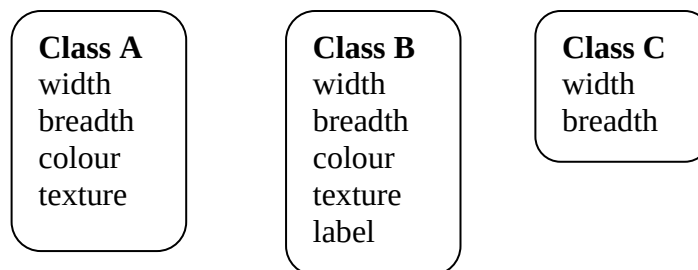
A typical list would look like this:

machine	operator	client	date
TLB	Khumalo	Smith	2009/09/23
TLB	Khumalo	Ramphele	2009/09/24
TLB	Khumalo	Kruger	2009/09/25
grinder	Tayob	Ramphele	2009/09/27

Write a SQL statement which will produce this list. (4)

Object-oriented programming

7.10 The following diagram shows three classes which are related to each other by inheritance. The diagram shows **only the fields (attributes)** of each of these three classes.



7.10.1 Which one of A, B or C is the ancestor class? (1)

7.10.2 Draw the inheritance hierarchy diagram for these classes. (3)

7.11 Every class has a constructor method.

7.11.1 Explain what happens when a constructor method is executed. (2)

7.11.2 What is a parameterised constructor? (1)

7.11.3 Give an example of a situation where it is necessary to use a parameterised constructor. (2)

7.12 Explain the difference between accessor methods and mutator methods. (2)

29 marks

Total: 180 marks